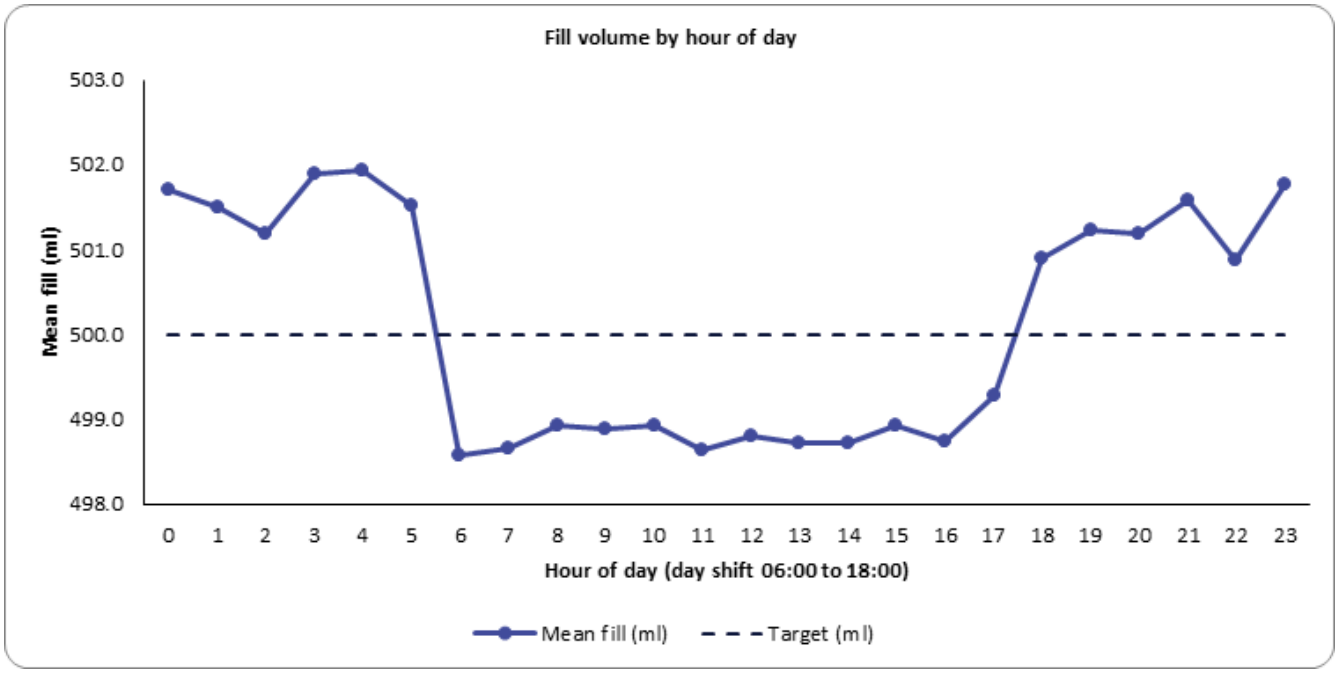
A02 worked exemplar: SPC control-chart workbook for the Lilongwe moringa fill line

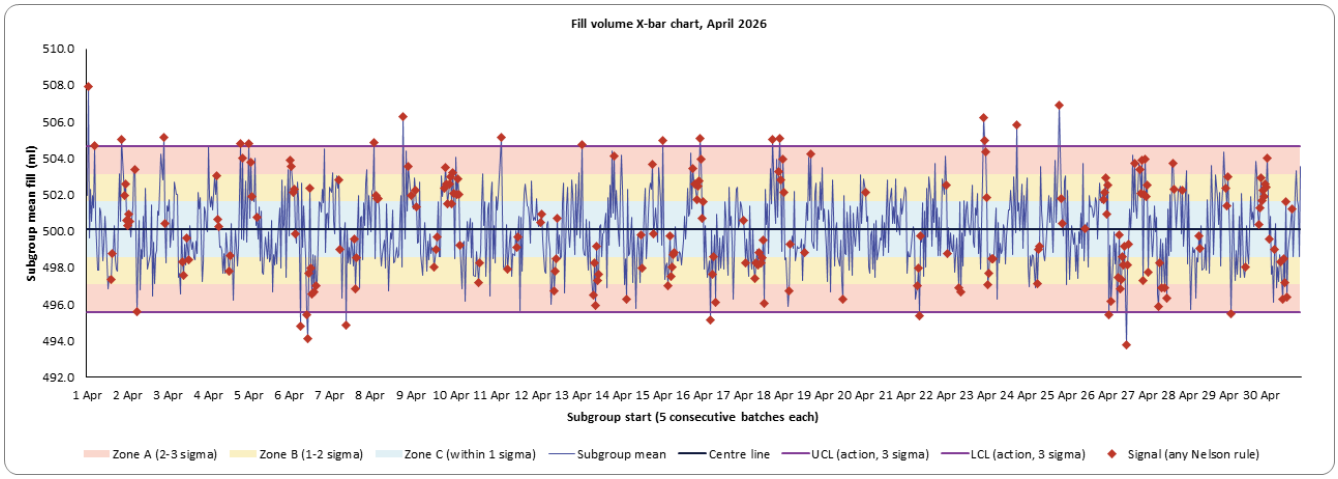
This is a worked example of an A02 submission, written by the programme team to show a workbook at the upper bands. It is set on Line 1 of Ubuntu Naturals' Lilongwe moringa plant, charting bottle fill volume from the W06 dataset, and not at the Johannesburg tableting line your brief describes. The two problems differ in kind. Lilongwe logs every batch around the clock and runs two shifts, so the work is choosing subgroups from a month of data; Johannesburg gives you 120 tablets already grouped, two characteristics and a press with punches and dies. Do not copy it. Its limits, signals, causes and OCAP belong to a liquid filler, so borrowed numbers or wording would answer the wrong process, and the Merit and Distinction bands are awarded for interpreting your own charts in tableting terms.

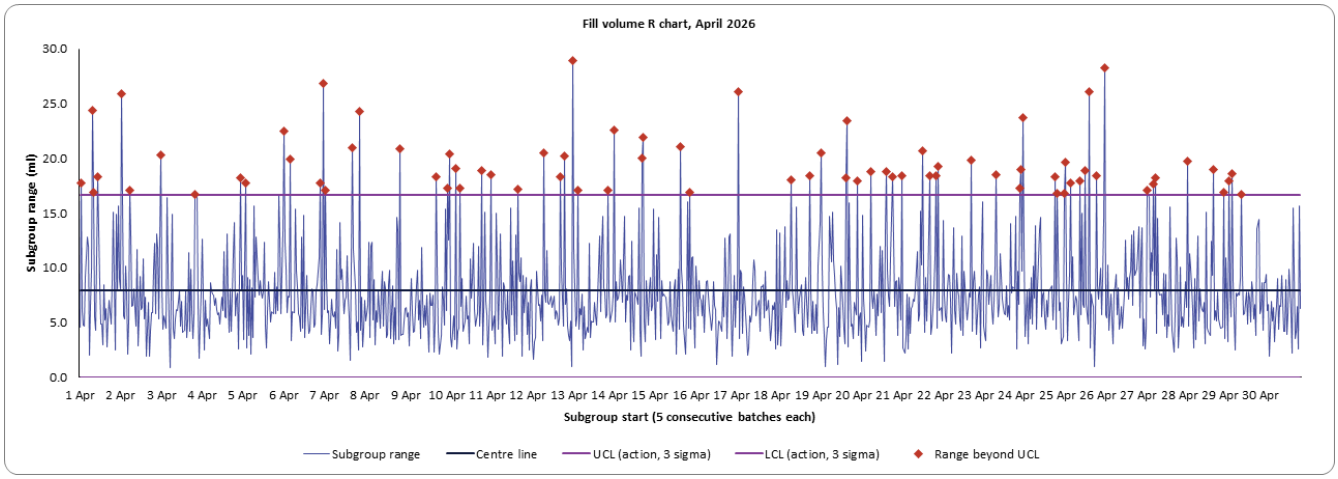
The submission

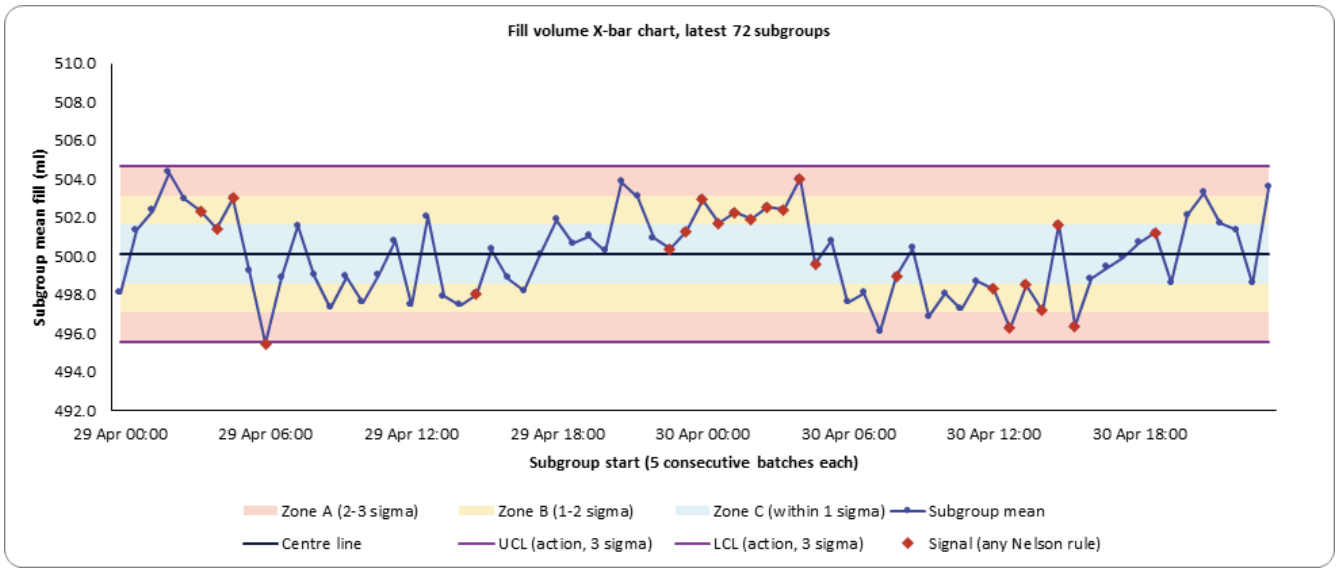
The submission is `A02_exemplar_lilongwe_moringa_SPC_Workbook.xlsx`, twelve labelled worksheets with a README sheet first. A learner's file would be named `A02_SPC_Workbook_[YourSurname].xlsx`; this one carries the exemplar name instead. The previews below are exported from Excel itself, so they show what Excel renders. The figures behind them are also in `A02_exemplar_lilongwe_moringa_figures.json`.

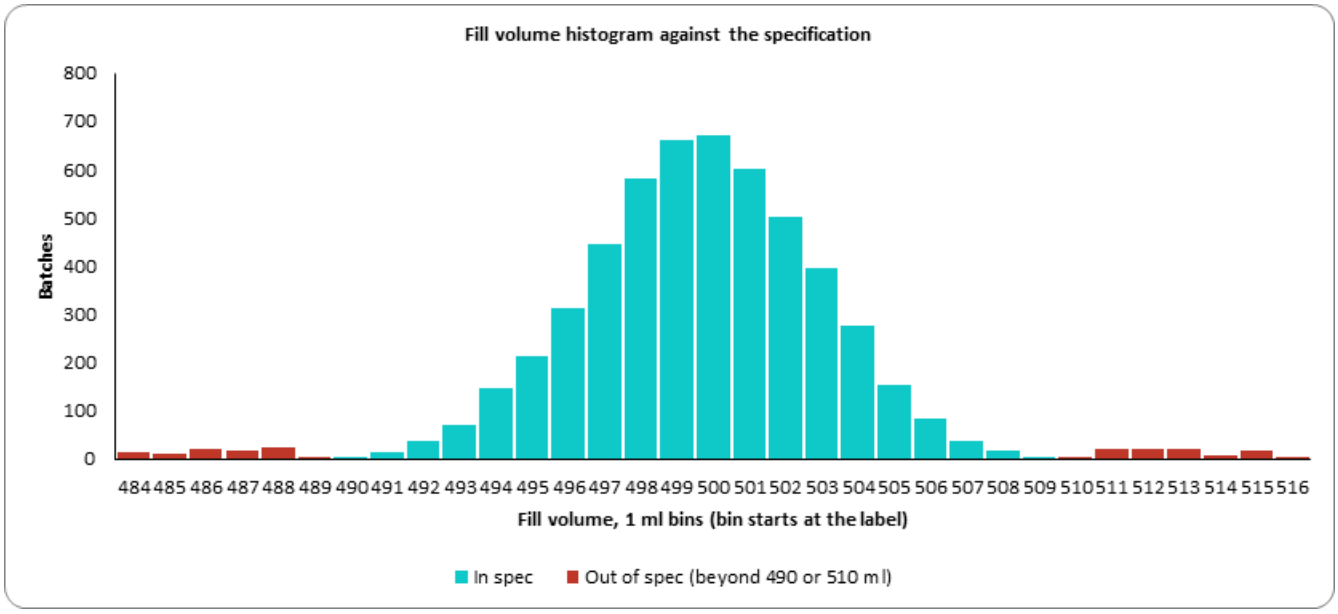


The hour-of-day profile is the finding in one picture. The fill steps down at 06:00 and up at 18:00 every day, so the month is two processes on one chart.


Fill volume X-bar chart, April 2026
Subgroup mean fill (ml)
510.0
508.0
506.0
504.0
502.0
500.0
498.0
496.0
494.0
492.0
1 Apr 2 Apr 3 Apr 4 Apr 5 Apr 6 Apr 7 Apr 8 Apr 9 Apr 10 Apr 11 Apr 12 Apr 13 Apr 14 Apr 15 Apr 16 Apr 17 Apr 18 Apr 19 Apr 20 Apr 21 Apr 22 Apr 23 Apr 24 Apr 25 Apr 26 Apr 27 Apr 28 Apr 29 Apr 30 Apr
Subgroup start (5 consecutive batches each)
Zone A (2-3 sigma)
Zone B (1-2 sigma)
Zone C (within 1 sigma)
Subgroup mean
Centre line
UCL (action, 3 sigma)
LCL (action, 3 sigma)
Signal (any Nelson rule)


Fill volume R chart, April 2026
Subgroup range (ml)
30.0
25.0
20.0
15.0
10.0
5.0
0.0
1 Apr 2 Apr 3 Apr 4 Apr 5 Apr 6 Apr 7 Apr 8 Apr 9 Apr 10 Apr 11 Apr 12 Apr 13 Apr 14 Apr 15 Apr 16 Apr 17 Apr 18 Apr 19 Apr 20 Apr 21 Apr 22 Apr 23 Apr 24 Apr 25 Apr 26 Apr 27 Apr 28 Apr 29 Apr 30 Apr
Subgroup start (5 consecutive batches each)
Subgroup range
Centre line
UCL (action, 3 sigma)
LCL (action, 3 sigma)
Range beyond UCL


Fill volume X-bar chart, latest 72 subgroups
Subgroup mean fill (ml)
510.0
508.0
506.0
504.0
502.0
500.0
498.0
496.0
494.0
492.0
29 Apr 00:00
29 Apr 06:00
29 Apr 12:00
29 Apr 18:00
30 Apr 00:00
30 Apr 06:00
30 Apr 12:00
30 Apr 18:00
Subgroup start (5 consecutive batches each)
Zone A (2-3 sigma)
Zone B (1-2 sigma)
Zone C (within 1 sigma)
Subgroup mean
Centre line
UCL (action, 3 sigma)
LCL (action, 3 sigma)
Signal (any Nelson rule)


Fill volume histogram against the specification
Batches
800
700
600
500
400
300
200
100
0
484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516
Fill volume, 1 ml bins (bin starts at the label)
In spec
Out of spec (beyond 490 or 510 ml)

OUT-OF-CONTROL ACTION PLAN

Lilongwe Line 1 filler: fill volume. Plot every 40 minutes.

Take the last 5 consecutive batches. Plot their mean on the X-bar chart and their range (largest minus smallest) on the R chart. Then work down the boxes.

Limits in force (from Limits sheet)	Value (ml)
X-bar action limits: UCL	504.7
X-bar centre line	500.1
X-bar action limits: LCL	495.6
X-bar warning lines (2 sigma)	497.1 and 503.2
R chart action limit: UCL	16.7
Specification	490 to 510

Check, in this order		Action
Q1. Is the RANGE above the R chart UCL?	**YES →**	YES: HOLD the filler. Re-measure the 5 bottles. Find the odd batch. Check that nozzle for drips or a part-block, foam on the bottle, bottle seating, and the level-probe reading. Reject or re-fill any bottle outside 490 to 510 ml. Call the shift supervisor.
NO ↓ go to the next question		
Q2. Is the MEAN above the UCL or below the LCL? (rule 1)	**YES →**	YES: STOP and call the shift supervisor. Compare the filler setpoint with the recipe sheet. Check product temperature at the filler and the supply-tank level. Restart only when the next subgroup plots inside the limits.
NO ↓ go to the next question		
Q3. 2 of the last 3 means beyond a warning line, 4 of the last 5 beyond 1 sigma, or 9 in a row on one side of the centre line? (rules 5, 6, 2)	**YES →**	YES: KEEP RUNNING. Check the setpoint against the recipe and ask what changed at the last changeover. Take the next subgroup early (20 minutes). A second signal this shift: call the shift supervisor.
NO ↓ go to the next question		
Q4. None of these?	→	Keep running. Next subgroup in 40 minutes.

Who is called, and how fast

Operator		Any signal: act on the box above within 5 minutes.
Shift supervisor		Q1 or Q2, or a second Q3 in one shift: at the line within 15 minutes; decides release or hold of the batches since the last good subgroup.
Quality engineer		Two supervisor calls in one shift, or an out-of-spec bottle found after release: same shift, within 4 hours; opens a deviation.
Operations Manager		The same signal on three shifts running: next working day; owns the corrective action.

Most likely causes on this filler

Wide range (Q1) — Nozzle drip or part-block; foaming at fill; bottle not seated; level-probe misread; one head out of step.

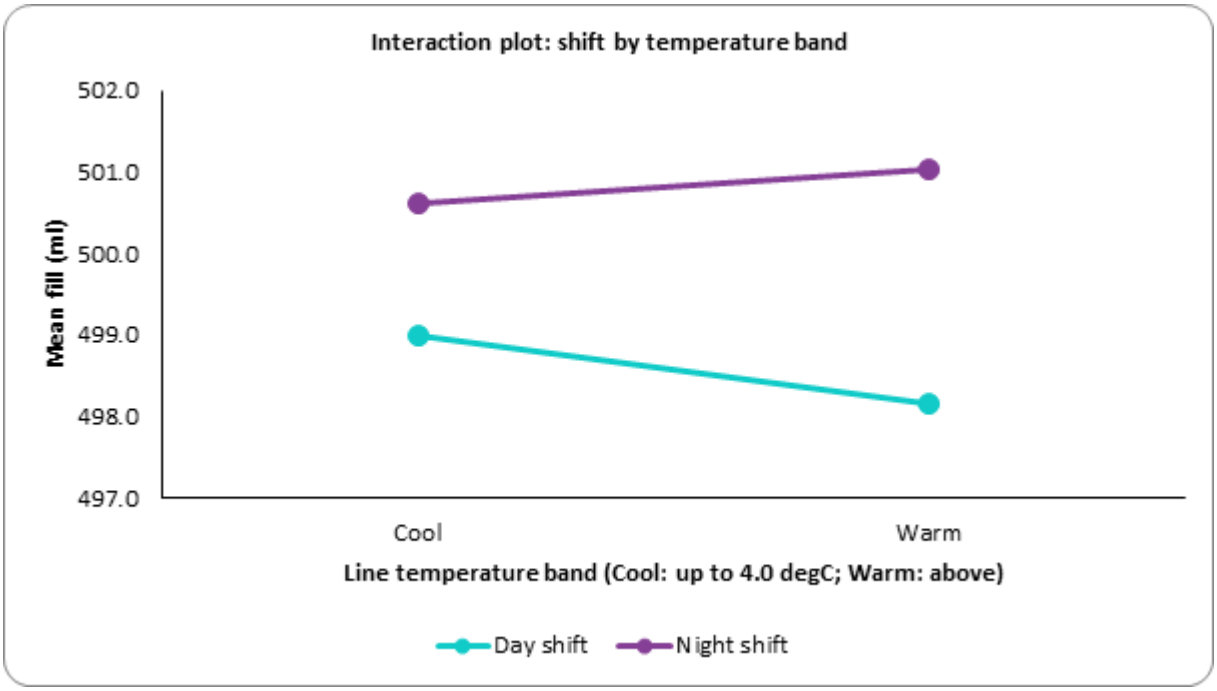
Mean jumps at a changeover (Q2, Q3) — Setpoint left at a different value by the outgoing crew; recipe not reloaded; product temperature at the filler different from the last shift.

Slow drift (rule 3) — Supply-tank level falling, product warming through the shift, wear in the dosing pump.

Write it down, every time

In the Signal_Log and the batch record: date, time, subgroup number, the question that said YES, what you checked, what you found, what you changed, the next subgroup's result, your initials. For any bottle outside 490 to 510 ml: how many, and where it went (re-fill, reject).

Review: the quality engineer reads the Signal_Log every week. Limits are recalculated only after a confirmed process change, from at least 25 new subgroups. This sheet is re-issued when the limits or the causes list change.



The workbook also has a line-temperature X-bar chart (`A02_exemplar_lilongwe_moringa_temperature_chart_preview.png`) and an R chart for the operator window.

Cleaning before charting

The CSV goes in untouched, with its 5,401 rows and the shift label typed 8 ways. Cleaning happens in columns beside it, so nothing is overwritten: the timestamp is split, the label is cleaned with TRIM, UPPER and LEFT and checked against the clock (no disagreements), the 1 duplicate row is flagged and excluded, and the energy entry in MWh is flagged, not corrected. The README sheet counts each of these with a formula, which is what a marker means by checking the data before charting it.

Choosing the subgroups

A subgroup is 5 consecutive batches from one shift, 40 minutes of filling. Five batches in a row share crew, setpoint, product and temperature, so the range inside a subgroup is short-term filler noise, and anything that changes between subgroups lands on the X-bar chart. A shift is 90 batches, so every shift divides into whole subgroups and the workbook confirms that all 1,080 subgroups come from one shift. Grouping across 06:00 or 18:00 would have put the shift change inside the ranges, inflated R-bar and widened every limit, hiding the main finding. The Commentary sheet explains why X-bar and R beats Individuals/MR and X-bar/S at this subgroup size.

Limits a marker can audit

The Limits sheet holds the standard constants table and looks up A2, D3, D4 and d2 for n. Every limit is a visible formula over the Subgroups sheet, written with named ranges so it reads as `=XBARBAR+c_A2*RBAR`. The grand mean is 500.12 ml and R-bar 7.89 ml, giving X-bar limits of 495.57 and 504.67 ml and an R-chart UCL of 16.68 ml. The within-subgroup sigma, R-bar over d2, is 3.39 ml. The charts plot the limit columns rather than drawn lines, shade the 1, 2 and 3 sigma zones, and mark signals with a separate series; conditional formatting picks out the same subgroups in the table.

Reading the signals

All eight Nelson rules are helper columns on every subgroup. The X-bar chart signals on 226 subgroups and the R chart on 76, 59 of them at night. Rule 6, four of five beyond one sigma, fires 123 times: the signature of a sustained shift of about one sigma. Day subgroups average 498.81 ml and night subgroups 501.43 ml, an offset of 2.62 ml, which is 1.7 times the standard deviation of a subgroup mean. The Signal_Log sheet lists every signalled subgroup with the rules that fired and a first hypothesis to check, and the Stratify sheet splits the line by shift and by hour, which is where the step at each changeover shows. The R-chart signals are different in kind: single batches far from the other four, the gross mis-fills.

Capability against performance

Over the month Cp is 0.98, Cpk 0.97, Pp 0.87 and Ppk 0.86, against a fill specification of 490 to 510 ml stated with the dataset. Cp and Cpk are nearly equal only because a low day and a high night average out near target; split by shift, Cpk is 0.93 on days and 0.79 on nights. The gap from Cp to Pp is the shift offset that the subgroups cannot see. The workbook gives intervals: with every batch the Cpk interval is 0.95 to 0.99, while 120 batches would give 0.83 to 1.11, too wide to tell whether the line reaches a Cpk of one. It recommends a Cpk of 1.33, which needs the within sigma cut to 2.51 ml. It also tests the normal model: Ppk predicts 9,330 PPM out of spec, the log shows 174 batches or 32,222 PPM, 3.5 times as many, because the gross mis-fills make the tails heavier than normal.

The OCAP

The OCAP sheet prints on one A4 page. It shows the limits in force (read live from the Limits sheet), four questions to answer in order with a yes action beside each, escalation by role with response times, the most likely causes on a filler, what to record and a weekly review. It uses the operator's words: hold, stop, keep running.

The experiment

The DoE sheet designs a replicated 2 x 2 trial: product temperature at the filler against the filler setup each crew leaves at changeover, 3 runs per cell, randomised, all on one day shift so that crew is held constant. No trial results exist, so its ANOVA block stays empty until they are typed in. The same two-way ANOVA formulas are rehearsed on the April log, shift against temperature band with 25 batches per cell: the shift effect is 2.25 ml and the temperature band is not significant ($p = 0.74$). That is why the trial varies setup and not only temperature.

Warning limits against action limits

For a stable process the month would give 2.9 false alarms at 3 sigma and 49.1 at 2 sigma. Against a shift the size of the night offset, 3-sigma limits take 9.9 subgroups on average to catch it and 2-sigma limits 2.6. The Commentary sheet turns that into a break-even: 2-sigma limits pay if an out-of-spec batch costs more than 1.7 times a wasted investigation. The costs are labelled assumptions in input cells; the recommendation (3 sigma to stop the line, 2 sigma to trigger an early re-check) holds across a wide range of them.

How it maps to the marking scheme

Rubric row	Band it aims at	Why	What would lift it a band
Chart Construction and Formulae	Distinction	Every limit is a visible formula on its own calculation sheet with named ranges and the constants table, the charts shade colour-coded sigma zones and mark signals, conditional formatting flags the same subgroups, and all eight Nelson rules are automated.	A template sheet the team could use from Monday, with the month charts on ranges that grow as rows are appended and a short operator training sheet; the window chart already follows new data.
Out-of-Control Analysis	Distinction	All eight rules are tested on every subgroup, the signals are tied to the filler (setpoint at changeover, nozzle, foam, level probe), and the stratification by shift and hour isolates the changeover step.	The log's hypotheses are generated from the rule pattern, so a few sentences repeat across 292 signals; Stretch would test each one against the batch record or maintenance log and record what was found.
Capability Indices	Distinction	All four indices are computed for the month and per shift, the Cp to Pp gap is explained as the shift offset, confidence intervals show what 120 batches could not tell you, and a target Cpk is recommended.	An improvement table showing Ppk and PPM as the offset is removed and sigma is cut step by step, benchmarked against a Six Sigma target.
OCAP Design	Distinction	It is one printable page in plain language, with a question-by-question decision ladder, escalation by role with response times, a cause checklist for a filler and recording and review rules.	Draw the decision ladder as a true flowchart and link every rule the workbook tests, including rules 4, 7 and 8, to its own cause checklist.
Design of Experiments (DoE)	Merit	The 2 x 2 design is complete, with levels, replication, randomised order, blocking and a power check, and the ANOVA is correct with effects and an interaction plot. It holds at Merit because the ANOVA with numbers in it is run on log data, where shift bundles crew, setup and temperature, not on the designed trial.	Run the 12 trial runs, or a short pilot of them, enter the results in the template and interpret those effects and their interaction for production practice.

Rubric row	Band it aims at	Why	What would lift it a band
Commentary and Risk Assessment	Distinction	The commentary links each chart pattern to process behaviour, costs the out-of-spec batches and the saving if nights matched days, and turns the 2-sigma against 3-sigma question into a break-even ratio with its assumptions in input cells.	A phased SPC roadmap from Line 1 to the plant, how the charts feed the quality system's deviation process, and how the night crew is trained to use them.

Where the figures come from

The builder, `scripts/build_a02_exemplar.py`, writes every figure as a live Excel formula over the raw CSV and computes the same figures itself for the commentary text and the sidecar JSON; `tests/test_exemplar_a02.py` recomputes each one independently and checks it against the workbook's own cached values. The CSV locators name the quantity the test computes from `lilongwe_moringa_30d.csv`. The cost figures, the trial's factor levels and the target Cpk are the author's choices and are not data.

Figure	Value	Source
Rows in the CSV	5,401	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:rows`
Duplicate rows excluded	1	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:duplicates`
Spellings of the shift label	8	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:shift_spellings`
Subgroups of 5 batches	1,080	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:subgroups`
Subgroups from one shift	1,080	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:one_shift`
Fill USL, ml	510	`workshops/06 excel to ai dashboarding/graduate/group assets/generate_factory_line_csv.py:USL_VOLUME`
Fill LSL, ml	490	`workshops/06 excel to ai dashboarding/graduate/group assets/generate_factory_line_csv.py:LSL_VOLUME`

Figure	Value	Source
Grand mean, ml	500.12	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:xbarbar`
R-bar, ml	7.89	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:rbar`
X-bar UCL, ml	504.67	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:ucl_x`
X-bar LCL, ml	495.57	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:lcl_x`
R chart UCL, ml	16.68	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:ucl_r`
Within-subgroup sigma, ml	3.39	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:sigma_w`
Day subgroup mean, ml	498.81	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:day_xbarbar`
Night subgroup mean, ml	501.43	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:night_xbarbar`
Night minus day, ml	2.62	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:offset`
Offset in sigmas of a subgroup mean	1.7	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:offset_sigma`
Subgroups with an X-bar signal	226	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:signals_x`
Subgroups with an R-chart signal	76	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:signals_r`
R-chart signals at night	59	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:signals_r_night`

Figure	Value	Source
Rule 6 signals	123	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:rule6`
Signalled subgroups in the log	292	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:signals_all`
Cp	0.98	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:cp`
Cpk	0.97	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:cpk`
Pp	0.87	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:pp`
Ppk	0.86	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:ppk`
Cpk, day shift	0.93	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:cpk_day`
Cpk, night shift	0.79	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:cpk_night`
Cpk 95% interval, lower, all batches	0.95	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:cpk_lo`
Cpk 95% interval, upper, all batches	0.99	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:cpk_hi`
Cpk 95% interval, lower, 120 batches	0.83	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:cpk_lo_120`
Cpk 95% interval, upper, 120 batches	1.11	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:cpk_hi_120`
Target Cpk (author's choice)	1.33	`scripts/build_a02_exemplar.py:CPK_TARGET`

Figure	Value	Source
Within sigma needed for the target, ml	2.51	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:sigma_required`
Out-of-spec batches	174	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:oos`
Observed PPM	32,222	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:ppm_observed`
Expected PPM under a normal model	9,330	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:ppm_expected`
Observed over expected	3.5	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:ppm_ratio`
False alarms a month, 3 sigma	2.9	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:fa3`
False alarms a month, 2 sigma	49.1	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:fa2`
Subgroups to catch the offset, 3 sigma	9.9	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/lilongwe_moringa_30d.csv:arl3`
Subgroups to catch the offset, 2 sigma	2.6	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:arl2`
Break-even cost ratio	1.7	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:breakeven`
Rehearsal ANOVA: shift effect, ml	2.25	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:anova_eff_shift`
Rehearsal ANOVA: temperature band p	0.74	`workshops/06 excel to ai dashboarding/graduate/group assets/lilongwe_moringa_30d.csv:anova_p_temp`